

Parker S. Ruth

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My research lies in the intersection of computing, engineering, and medicine. I'm currently designing sensors and algorithms to measure digital biomarkers of cardiovascular and neuromuscular health. I am fortunate to work closely with mentors and collaborators in computer science, statistics, bioengineering, and medicine.

Education

Stanford University	PhD, Computer Science	2021 – 2027
Stanford University	MS, Computer Science	2021 – 2024
Fellowships: NSF GRFP, NIH F31		
Thesis Committee: James Landay, Scott Delp, Todd Coleman, Emily Fox, Alison Marsden		
University of Washington	BS, Bioengineering	2016 – 2021
University of Washington	BS, Computer Engineering	2016 – 2021
College Honors, <i>summa cum laude</i> GPA 3.96		
Research Advisor: Shwetak Patel		

Publications and Invited Talks

Under Review

- [1] Hannah Heigold, **Parker S. Ruth**, Julie Muccini, Dawit Lee, Sydney Barta, Ariana Rodrigues, Trevor Hastie, Kristen K. Steenerson, and Scott L. Delp. **StaBLE: Digital metrics capture balance performance across a wide spectrum of balance tasks and abilities.** *bioRxiv*
[10.64898/2026.09.16.750744](https://doi.org/10.64898/2026.09.16.750744)
- [2] Sydney C. Covitz, **Parker S. Ruth**, Shelby Vogt-Domke, Carmichael Ong, Tian Tan, Audrey Chun, Sarah Ismail, Lin Karman, Julie Muccini, Shannon Li, Melina Rogers, Scott Uhlich, John W. Day, Jennifer L. Hicks, prefix=de useprefix=false family=Monts, given=Constance, Tina Duong, and Scott L. Delp. **Deep learning models detect neuromuscular disease and quantify functional impairment from video-derived biomechanics data.** *bioRxiv*
[10.64898/2026.09.30.754736](https://doi.org/10.64898/2026.09.30.754736)

Peer-Reviewed Publications

- [3] **Parker S. Ruth**, Tommy DeBenedetti, Lily O'Brien, James A. Landay, Todd Coleman, and Emily B. Fox. **Vascular waveform analysis using Bayesian pulse deconvolution.** *npj Digital Medicine*, (to appear), August 2026
[10.64898/2026.02.09.699383](https://doi.org/10.64898/2026.02.09.699383)
- [4] Basma Altaf, Arash Tavakoli, **Parker S. Ruth**, Andrea Green, Jiaxuan Xu, Sneha Jain, Ethan Chiu, Lucy Zhang Bencharit, Elizabeth L. Murnane, James A. Landay, and Sarah L. Billington. **Impact of windows, natural materials, and diverse representations in built environments on psychological and physiological well-being: A between-subjects experiment in immersive virtual environments.** *Building and Environment*, 280:113147. ISSN 0360-1323
[10.1016/j.buildenv.2025.113147](https://doi.org/10.1016/j.buildenv.2025.113147)
- [5] **Parker S. Ruth**, Scott D. Uhlich, Constance de Monts, Antoine Falisse, Julie Muccini, Sydney Covitz, Shelby Vogt-Domke, John Day, Tina Duong, and Scott L. Delp. **Video-Based Biomechanical Analysis Captures Disease-Specific Movement Signatures of Different Neuromuscular Diseases.** *New England Journal of Medicine Artificial Intelligence (NEJM AI)*, 2(9), August 2025
[10.1056/AIoA2401137](https://doi.org/10.1056/AIoA2401137)

- [6] Alvin Cao, Ken Christofferson, **Parker S. Ruth**, Naveed Rabbani, Yuanchun Shi, Alex Mariakakis, Yuntao Wang, and Shwetak Patel. **EarSteth: Cardiac Auscultation Audio Reconstruction Using Earbuds**. *46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, pages 1–4, July 2024
[10.1109/EMBC53108.2024.10781641](https://doi.org/10.1109/EMBC53108.2024.10781641)
- [7] Jason S. Hoffman, Matthew Hirano, Nuttada Panpradist, Joseph Breda, **Parker S. Ruth**, Yuanyi Xu, Jonathan Lester, Bichlien H. Nguyen, Luis Ceze, and Shwetak N. Patel. **Passively sensing SARS-CoV-2 RNA in public transit buses**. *Science of The Total Environment*, 821:152790, May 2022. ISSN 0048-9697
[10.1016/j.scitotenv.2021.152790](https://doi.org/10.1016/j.scitotenv.2021.152790)
- [8] Justin D. Vrana, Nuttada Panpradist, Nikki Higa, Daisy Ko, **Parker S. Ruth**, Ruth Kanthula, James J. Lai, Yaoyu Yang, Samar R. Sakr, Bhavna Chohan, Michael H. Chung, Lisa M. Frenkel, Barry R. Lutz, Eric Klavins, and Ingrid A. Beck. **Implementation of an interactive mobile application to pilot a rapid assay to detect HIV drug resistance mutations in Kenya**. *PLOS Global Public Health*, 2(2):e0000185, February 2022. ISSN 2767-3375
[10.1371/journal.pgph.0000185](https://doi.org/10.1371/journal.pgph.0000185)
- [9] Jackson J. Wallner, Ingrid A. Beck, Nuttada Panpradist, **Parker S. Ruth**, Humberto Valenzuela-Ponce, Maribel Soto-Nava, Santiago Ávila-Ríos, Barry R. Lutz, and Lisa M. Frenkel. **Rapid Near Point-of-Care Assay for HLA-B*57:01 Genotype Associated with Severe Hypersensitivity Reaction to Abacavir**. *AIDS Research and Human Retroviruses*, 37(12):930–935, December 2021. ISSN 0889-2229
[10.1089/aid.2021.0103](https://doi.org/10.1089/aid.2021.0103)
- [10] Nuttada Panpradist, Qin Wang, **Parker S. Ruth**, Jack H. Kotnik, Amy K. Oreskovic, Abraham Miller, Samuel W. A. Stewart, Justin Vrana, Peter D. Han, Ingrid A. Beck, Lea M. Starita, Lisa M. Frenkel, and Barry R. Lutz. **Simpler and faster Covid-19 testing: Strategies to streamline SARS-CoV-2 molecular assays**. *eBioMedicine*, 64:103236, February 2021. ISSN 2352-3964
[10.1016/j.ebiom.2021.103236](https://doi.org/10.1016/j.ebiom.2021.103236)
- [11] **Parker S. Ruth**, Jerry Cao, Millicent Li, Jacob E. Sunshine, Edward J. Wang, and Shwetak N. Patel. **Multi-Channel Facial Photoplethysmography Sensing**. *42nd Annual International Conference of the IEEE Engineering in Medicine Biology Society (EMBC)*, pages 4179–4182, July 2020
[10.1109/EMBC44109.2020.9176700](https://doi.org/10.1109/EMBC44109.2020.9176700)
- [12] Nuttada Panpradist, Ingrid A. Beck, **Parker S. Ruth**, Santiago Ávila-Ríos, Claudia García-Morales, Maribel Soto-Nava, Daniela Tapia-Trejo, Margarita Matías-Florentino, Hector E. Paz-Juarez, Silvia del Arenal-Sanchez, Gustavo Reyes-Terán, Barry R. Lutz, and Lisa M. Frenkel. **Near point-of-care, point-mutation test to detect drug resistance in HIV-1: A validation study in a Mexican cohort**. *AIDS*, 34(9):1331–1338, July 2020. ISSN 0269-9370
[10.1097/QAD.0000000000002524](https://doi.org/10.1097/QAD.0000000000002524)
- [13] Nuttada Panpradist, Ingrid A. Beck, Justin Vrana, Nikki Higa, David McIntyre, **Parker S. Ruth**, Isaac So, Enos C. Kline, Ross Milne, Ruth Kanthula, Annie Wong-On-Wing, Jonathan Lim, Daisy Ko, Theresa Rossouw, Ute D. Feucht, Michael Chung, Gonzague Jourdain, Nicole Ngo-Giang-Huong, Laddawan Laomanit, Jaime Soria, James Lai, Eric E. Klavins, Lisa M. Frenkel, and Barry R. Lutz. **OLA-Simple: a software-guided HIV-1 drug resistance test for low-resource laboratories**. *eBioMedicine*, 50:34–44, December 2019. ISSN 2352-3964
[10.1016/j.ebiom.2019.11.002](https://doi.org/10.1016/j.ebiom.2019.11.002)

Conference Posters and Abstracts

- [14] **Parker S. Ruth**, Alisa Hathaway, Chloe Zhong, Milly Wong, Zack Martino, John Li, James Landay, Jeremy Dahl, and Coleman Todd P. **Wearable ultrasound for cardiovascular digital twins**. *Neuromuscular Disorders*, 53:105735. ISSN 0960-8966
[10.1016/j.nmd.2025.105735](https://doi.org/10.1016/j.nmd.2025.105735)

- [15] C. De Monts De Savasse, **Parker S. Ruth**, S. Ulrich, S. Vogt-Domke, S. Ismail, L. Karman, A. Falisse, J. Muccini, S. Covitz, J. Day, S. Delp, and T. Duong. **Video-based biomechanical analysis captures disease-specific movement signatures of myotonic dystrophy and facioscapulohumeral muscular dystrophy**. *Neuromuscular Disorders*, 53:105735. ISSN 0960-8966
[10.1016/j.nmd.2025.105735](https://doi.org/10.1016/j.nmd.2025.105735)
- [16] C. De Monts De Savasse, **Parker S. Ruth**, S. Ulrich, S. Vogt-Domke, S. Ismail, L. Karman, A. Falisse, J. Muccini, S. Covitz, J. Day, S. Delp, and T. Duong. **Towards Video-Based Movement Biomarkers for Neuromuscular Diseases**. *Converging Clinical and Engineering Research on Neurorehabilitation V*, pages 501–504
[10.1007/978-3-031-77584-0_98](https://doi.org/10.1007/978-3-031-77584-0_98)
- [17] **Parker S. Ruth**, Constance de Monts, Scott Uhrich, Julie Muccini, Paxton Ataide, Antoine Falisse, John Day, Scott Delp, and Tina Duong. **Digital Movement Biomarkers for Neuromuscular Diseases from Smartphone Videos**. *Myotonic Dystrophy Foundation Annual Conference*, September 2023

Invited Talks

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| [1] Vascular waveform analysis using Bayesian pulse deconvolution
Duke University Center for Computational and Digital Health Innovation | 10/2026 |
| [2] Accelerating Biomarker Discovery through Large-Scale Motion Capture Using OpenCap
Stanford University FAIR Center Webinar (webinar link) | 6/2026 |
| [3] Towards Smartphone Video-Based Biomarkers of Human Movement
University of Washington Ubiquitous Computing Seminar | 3/2025 |
| [4] Towards Smartphone Video-Based Biomarkers of Human Movement
University of California San Diego Design Lab Meeting | 2/2025 |
| [5] Scalable Kinematic Analysis Using Smartphone Videos: Towards Movement Biomarkers for Neuromuscular Diseases
MR3 Network 2023 Scientific Retreat | 9/2023 |
| [6] Multi-Channel Facial Photoplethysmography Sensing
42nd Annual International Conferences of the IEEE Engineering in Medicine and Biology Society (EMBC) | 7/2020 |

Teaching Experience

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| TA, CS 109 Probability for Computer Scientists
• Led Oxford-style tutorials teaching foundational probability concepts | 9/2026 – 12/2026 |
| Program Facilitator, AI in Medicine High School Summer Program
• Designed and led 2-week program with guest speakers, lab tours, and project work
• Delivered workshops teaching biomechanics, computing, design, and AI for medicine | 6/2026 – 7/2026 |
| TA, CS 347 Human-Computer Interaction: Foundations and Frontiers
• Led weekly discussion sections on seminal literature in human-computer interaction
• Wrote quizzes and graded reading reflection assignments
• Lectured on Human-Computer Interaction and Health
• Course evaluation teaching effectiveness 4.81 / 5.00 | 1/2024 – 3/2024 |
| Instructor, CSE 590U Ubiquitous Computing Graduate Seminar
• Led weekly discussion section with guest presenters and paper critique
• Topics included interaction techniques, wearables, novel sensing, and pervasive computing | 9/2019 – 6/2020 |
| Co-instructor, BIOEN 217 MATLAB Fundamentals For Bioengineers
• Co-instructed seminar introducing programming in MATLAB with biomedically relevant examples
• Prepared and delivered lectures, graded coding assignments, and supported course development | 9/2019 – 12/2019 |

Author, Biosignal Processing Course Text

8/2018 – 9/2020

- Wrote [140-page course text](#) for UW bioengineering sensors course
- Introduces signal acquisition, Fourier analysis, and digital and analog filters

Mentoring

• Jason Alonso (Stanford University)	6/2026 – Present
• John Li (Stanford University)	6/2026 – Present
• Sophia Rong (Stanford University)	6/2026 – Present
• Sonam Ngamdung (Stanford University)	6/2026 – 8/2026
• Andy Phuong (Stanford University)	6/2026 – 8/2026
• Davido Zhang (Stanford University)	3/2026 – 7/2026
• Sohan Chunduru (Stanford University)	1/2026 – 7/2026
• Indu Gadiraju (Stanford University)	1/2026 – Present
• Zack Martino (West Valley College)	9/2025 – 6/2026
• Lily O'Brien (Stanford University)	6/2025 – Present
• Chloe Zhong (Stanford University)	3/2025 – Present
• Milly Wong (Stanford University)	3/2025 – Present
• Nathalie Moreno (Stanford University)	10/2024 – 12/2025
• Ron Polonsky (Stanford University)	3/2025 – 6/2025
• Eli Waldman (Stanford University)	3/2025 – 6/2025
• Amanda Phan (Stanford University)	1/2025 – 6/2026
• Tommy DeBenedetti (Stanford University)	6/2024 – 3/2026
• Morayo Adeyemi (Howard University)	6/2024 – 9/2024
• Jordan Rodriguez (University of Arizona)	6/2024 – 9/2024
• Ege Turan (Stanford University)	10/2023 – 12/2023
• Alexandra Collins (Stanford University)	6/2023 – 9/2023
• Hamad Musa (Stanford University)	6/2023 – 9/2023

Service

Peer Reviewing

• NIH-Funded Stanford FAIR Center pilot project proposals	05/2026
• Journal of Shoulder and Elbow Surgery	10/2025
• Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies	3/2025
• ACM CHI conference on Human Factors in Computing Systems, late-breaking work	2/2024

Stanford Dean's Graduate Student Advisory Council

9/2024 – 6/2025

- Led initiative to implement individual development plans across School of Engineering
- Created first School of Engineering student experience feedback survey
- Fostered inter-department social connections through event funding

Volunteering and Outreach

• Presenter, Stanford First Year Research Seminar	02/2026
• Reviewer, Stanford Computer Science Faculty Search Committee	12/2025
• Presenter, Stanford EXPLORE High School Outreach Program	7/2025
• Demonstrator, Stanford School of Engineering Centennial Showcase	5/2025
• Workshop Mentor, OpenSim+ Advanced Workshop, Neuromuscular Biomechanics Lab	3/2025, 3/2026
• Workshop Facilitator, Co-design for Healthcare and Assistive Technology	3/2025
• Volunteer, Stanford Computer Science Application Support Program	12/2024
• Workshop Facilitator, LINXS Computer Science Outreach Research Program	7/2024

- Reviewer, Stanford Computer Science PhD Admissions Committee 12/2022, 12/2023
- Volunteer, Stanford Computer Science Application Support Program 12/2023
- Presenter, UW Computer Science CS4Teachers outreach event 7/2019
- Volunteer, UW Engineering Discovery Days 4/2018, 4/2019
- Representative, UW Bioengineering Department Curriculum Committee

Awards and Honors

Grants & Fellowships

NIH F31 Predoctoral Fellowship, \$148,000	2025
Wu Tsai Human Performance Alliance Seed Grant, \$200,000	2024
Tau Beta Pi Fellowship, \$10,000	2021
National Science Foundation Graduate Research Fellowship, \$138,000	2021

National Awards and Honors

Hertz Fellowship Finalist	2022
CRA Outstanding Undergraduate Researcher Award Finalist	2020, 2021
Barry Goldwater Scholarship	2020
Davidson Fellows Scholarship Honorable Mention	2016
National Merit Scholarship	2016

Undergraduate Awards and Honors

Paul G. Allen School Outstanding Senior Award	2021
Paul G. Allen School Best Senior Thesis Award	2021
College of Engineering Dean's Medal for Academic Excellence	2021
Annual Dean's List	2017 – 2020
Husky 100 Award	2020
Mary Gates Research Scholarship	2018, 2020
Levinson Emerging Scholars Award	2019
Microsoft Endowment Scholarship	2019
Patricia G. Lynch and Theodora & Eugene Russell Memorial Scholarship	2019
Tau Beta Pi Engineering Honors Society	2018
Washington Research Foundation Fellowship	2018
Mary Gates Leadership Scholarship	2018
Mary Gates Achievement Scholarship	2017